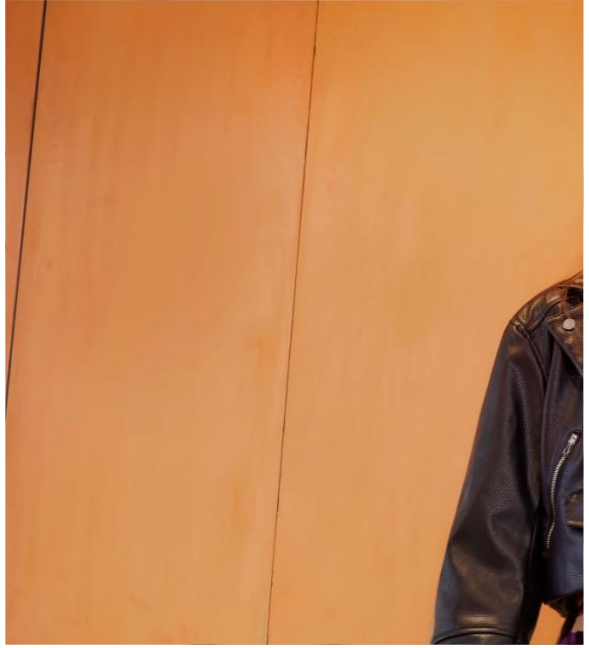






Elevator Energy Calculation Report

Project Name

- Best-in-class energy efficiency according to ISO 25745
- Smart and green building design
- Externally assured product declarations
- Durability and sustainability of materials and solutions

24 SEPTEMBER 2026

Dedicated to People Flow™

General Information

Customer Contact Information

Client	
Project	
Site	Bahrain

Main Elevator Specification

Platform Type	KONE N MonoSpace 21.2
Quantity of identical elevators	1
Group Name	

KONE Contact Information

Contact first name	Gokhan
Contact last name	Uzumcu
Address	
City	
Country	Bahrain
Phone	
Email	gokhan.uzumcu@kone.com



Energy performance

SINGLE UNIT ELEVATOR ANALYSIS ACCORDING TO ISO 25745-2

Energy Savings

55%

ISO 25745-2 energy efficiency class (expected)

B

Calculated annual expenditure for electricity(1) [BHD]

47

CO_2e emissions Operational CO_2e [$kgCO_2e$/Lifetime]

21619

ISO 25745-2 annual energy consumption [kWh/year]

1572

Annual energy savings by regenerative drive.

20%

(1) The calculation is based on the expected ISO 25745 annual energy consumption, using a kWh price of BHD 0.03 per kWh.

The results are KONEs best estimates. The real-life values may vary depending on the actual product specification, delivery or installation. Therefore, KONE does not give any representations or warranties that these calculations are the real-life values and shall not be liable in connection with the use of these calculations. Calculations provided are intended solely for the internal use of the recipient organization(s). They must not be shared externally, reused, or commercialized in any form, including to or by third-party consultants or advisors engaged by the recipient. Any reproduction, distribution, or use beyond the original purpose requires prior written consent.





Energy Performance Calculation Results

EXISTING TECHNOLOGY COMPARISON WITH THE PROPOSED MODERNIZATION PACKAGE

	Annual energy consumption(1)	Annual Regenerative Energy	Annual expenditure for electricity	Lifetime expenditure for electricity	Operational CO_2e	Embodied CO_2e
Elevator Technology	**[kWh/year]**	**[kWh/year]**	**[BHD/year]**	**[BHD/lifetime]**	**[$kgCO_2e$/lifetime 20 years]**	**[$kgCO_2e$]**
Typical Traction Elevator	3524	N/A	106	2576	48498	0
KONE N MonoSpace (FRB)	1572	382	47	1145	21619	15000

55% Energy Savings

(1) Calculated with ISO 25745-2:2015/Amd.1:2023 & ISO 25745-1:2023

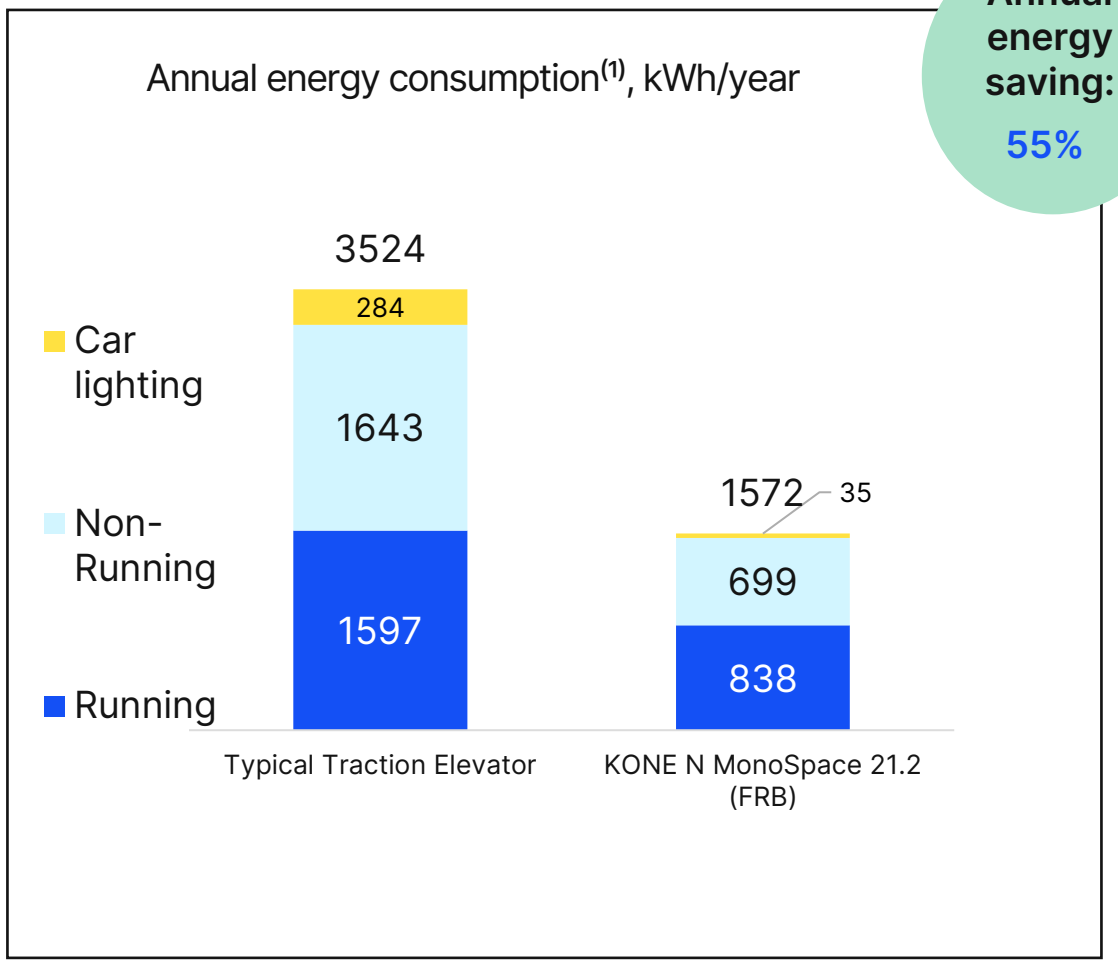
(2) "Operational" CO_2e refers to emissions from the elevator's energy use over its lifetime. "Embodied" CO_2e refers to emissions from materials, manufacturing, transport, and installation (A1–A5)

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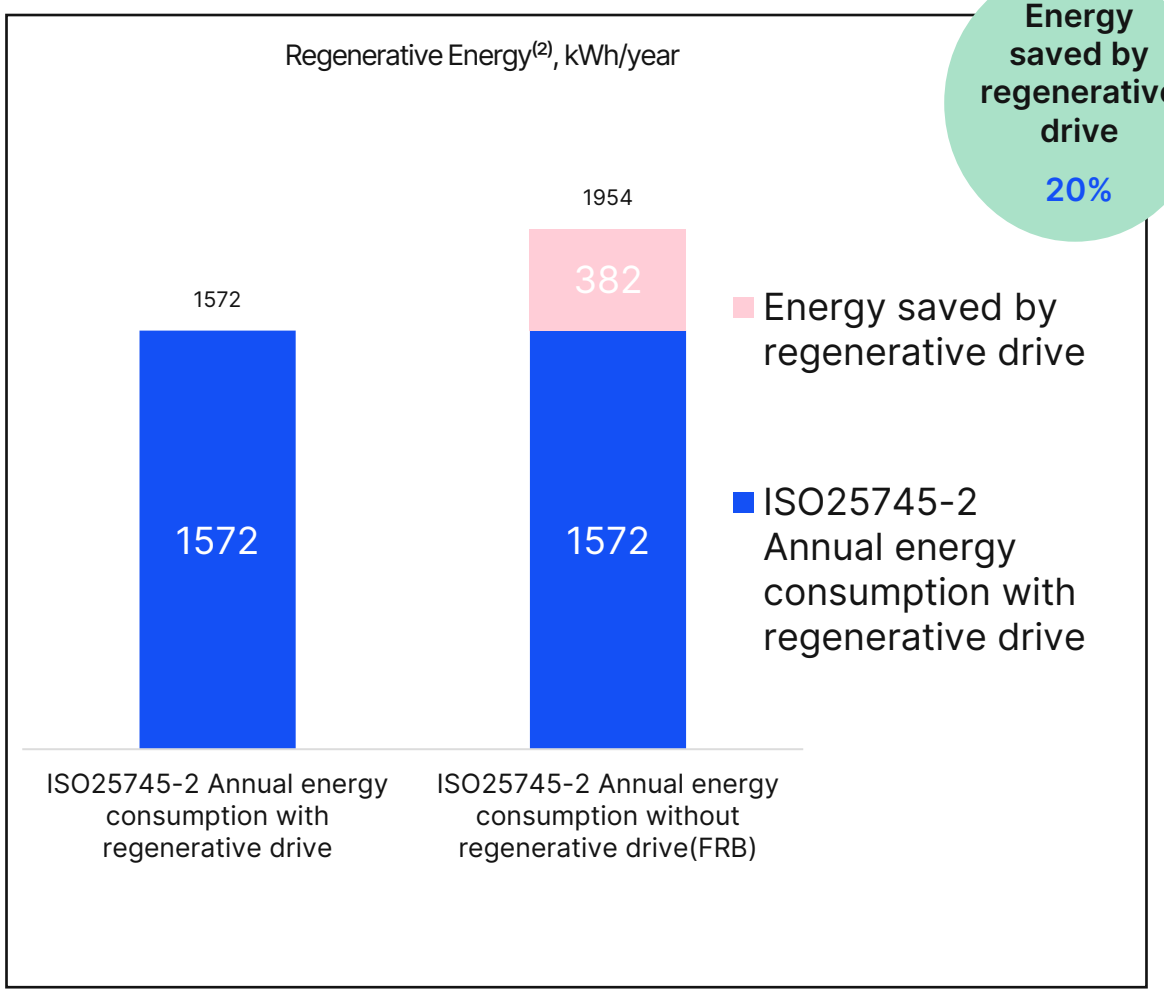
Energy Savings

ANNUAL ENERGY CONSUMPTION AND REGENERATIVE ENERGY

Annual energy saving: 55%



Energy saved by regenerative drive 20%



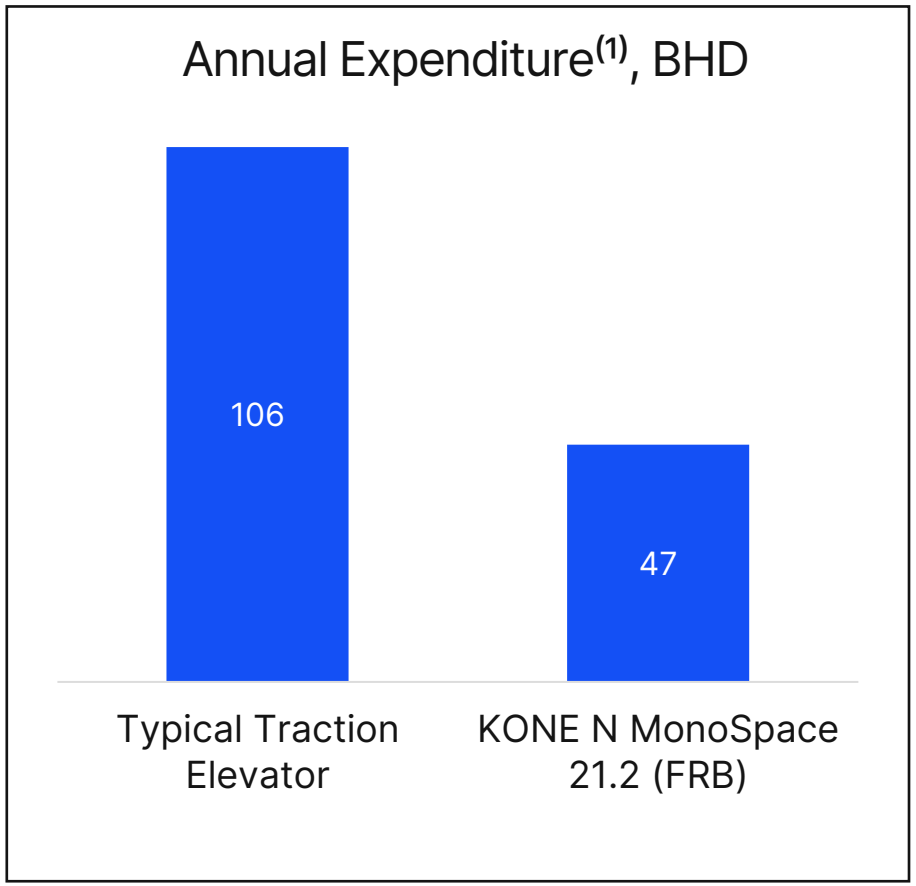
(1) Annual energy consumption is the total of the energy consumed while the elevator is moving and the consumption from electrical devices (e.g., lighting, signalization).

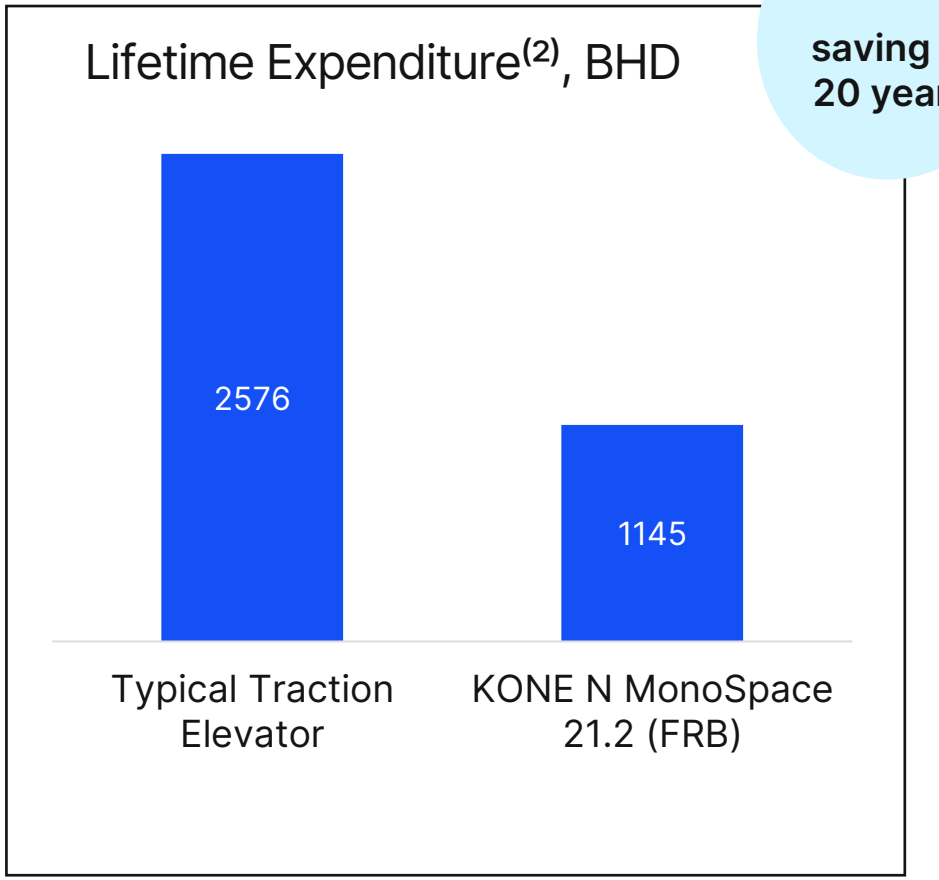
(2) The regenerative energy corresponds to the energy that is produced by the elevator thanks to the regenerative drive. The electricity generated by the regenerative drive can be fed back to the building for other uses.

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Expenditure savings

ANNUAL AND LIFETIME EXPENDITURE FOR ELECTRICITY





1k BHD

saving in 20 years

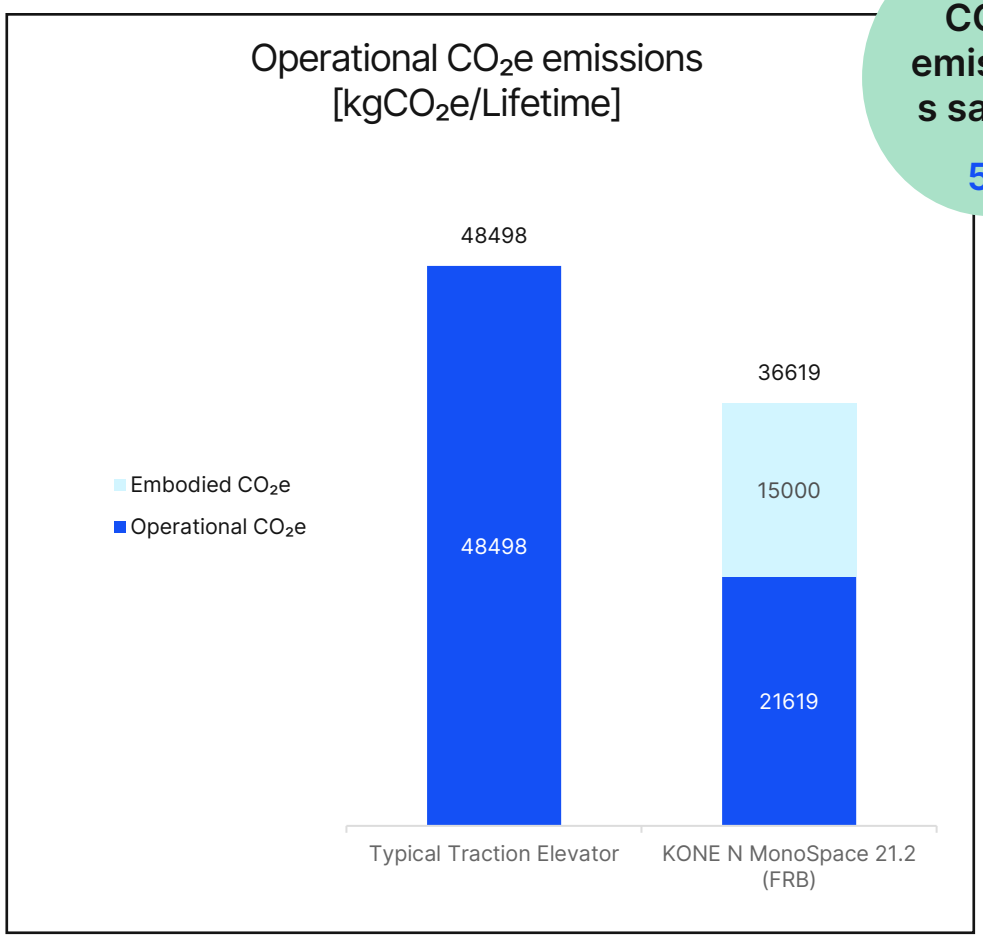
(1) Annual expenditure is the amount paid by the customer for the electric bill required to operate each elevator.

(2) Lifetime expenditure is the money paid by the customer in elevator lifetime for the electricity bill.

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Operational CO_2e emissions

The elevator's operational CO_2e emissions are determined by its electricity consumption, and the carbon intensity of the local electricity generation mix at the installation site.



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Summary of Input Data

Summary of Input Data	As existing technology	As modernization technology
Nominal load [kg]	1600	1600
Nominal speed [m/s]	1	1
Travel distance [m]	5.25	
Lifetime [year]	20	
Number of served floors	2	
Starts per year	109500	
Operating days per year	365	
Inflation rate [%]	2	
Country	Bahrain	
Electricity price [BHD/kWh]	0.03	
ISO25745 Usage category	3	
Gear	Gearless	
Electric motor type	Alternate current (AC)	
Drive type	V3F PMSM	
Signalization type	LED	
Machine room location	PW	
Controller type	Microprocessor	
Roping suspension type	2:1	
Rope wrap type	CSW	
Ceiling type	Fluorescent Tubes	CL80
OCL	OCL	OCL
Regenerative drive	No	Yes
Car Fan	No	No

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Summary of Input Data

Summary of Input Data	
Flexible Car	Yes
No of Elevators in Group	Simplex
LIL	LIL AMB
Number of COP	1
CH	2200
Control system	KCE
No of floors with LCI	2
IntegrateLCI	FCB
ExistingOCL	Yes
Modern OCL	Yes
Modern Regenerative Drive	Yes
ACU F	Yes
EBD A	Yes
Display Types	Dot Matrix
ABE M	Yes
KFC	Yes
Number of landings	2
No of car entrances	1
FRD	Yes
Door Type	middle duty D15
KRM	KRM G

Summary of Input Data	
Group Control	Simplex
Car width [mm]	1500
Car depth	2200
Quantity of identical elevators	1
Running time [h / operating day]	1.79

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ISO 25745-2 Energy performance

SINGLE UNIT ELEVATOR ANALYSIS – DETAILED RESULTS

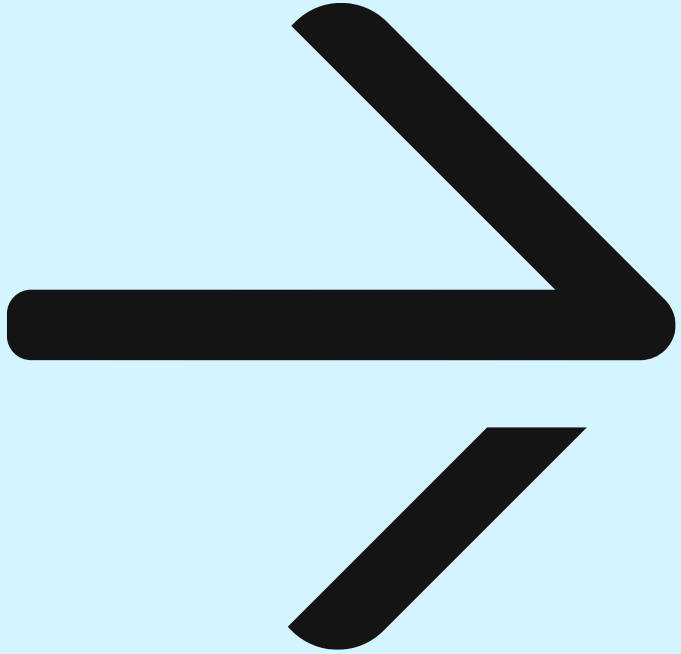
Specific usage category	ISO 25745-2 energy efficiency class (expected)	Idle performance level	$Standby_5$ performance level	$Standby_{30}$ performance level	Running performance level
3	B	3	2	2	2

Operating days per year	ISO 25745-2 annual energy consumption [kWh/year]	Calculated annual expenditure for electricity(1) [BHD]	Energy saved by regenerative drive [kWh/year]	Annual energy consumption % saved by regenerative drive
365	1572	47	382	20%

(1) The calculation is based on the expected ISO 25745 annual energy consumption, using a kWh price of BHD 0.03 per kWh.

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Appendix



Usage categories and number of trips per day

AS PER ISO25745-2

Usage category	1	2	3	4	5	6
Usage intensity/frequency	Very low	Low	Medium	High	Very High	Extremely High
Number of trips per day (nd) Typical range	50 (<75)	125 (75 to <200)	300 (200 to <500)	750 (500 to <1000)	1500 (1000 to <2000)	2500 (≥2000)
Typical buildings and usage (Operating days per year)	Residential building up to 6 dwellings (360 d) Residential care home (360 d) Small office or administrative building with few operations (260 d) Suburban railway stations (360 d)	Residential building up to 20 dwellings (360 d) Small office or administrative building with 2 to 5 floors (260 d) Small hotels (360d) Office Car parks (260 d) General Car parks (360 d) Main line railway stations (360 d) Library (312 d) Entertainment Centres (360d) Stadia (intermittent)	Residential building with up to 50 dwellings (360 d) Medium-sized office or administrative building with up to 10 floors (260 d) Medium-sized hotel (360 d) Airports (360 d) University (260 d) Small hospital (360 d) Shopping Centre (360 d)	Residential building with more than 50 dwellings (360 d) Large office or administrative building with more then 10 floors (260 d) Large hotel (360 d)	Very large office or administrative building over 100 m height (260 d)	Very large office or administrative building over 100 m height (260 d)
Typical rated speed	0,63 [m/s]	1,00 [m/s]	1,60 [m/s]	2,50 [m/s]	5,00 [m/s]	5,00 [m/s]

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ISO 25745-2: Performance levels and overall classification

- ISO 25745-2 standard is divided into 7 performance levels for running and idle/standby$_5$/standby$_{30}$
- Energy efficiency classification thresholds are a combination of running and idle/standby$_5$/standby$_{30}$ performances

- The overall energy efficiency class is based on evaluating the total estimated daily energy consumption against the thresholds of the classes, which also depend on the rated load and number of trips per day used in the calculation

- Reaching the best total energy efficiency class is more important than reaching the best performance levels in running or in idle/standby

ISO 25745-2 performance levels for idle and standby



ISO 25745-2 performance levels for running



ISO 25745-2 energy efficiency class (expected)



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You can read more about KONE sustainability focus areas here